

CO4- AT2 - Lab Practice

Java Lab Practice – Applets, AWT Components and Layout Managers

I. MD IMTHIAZ

192521360

```
import java.awt.*;
```

```
import java.awt.event.*;
```

```
public class Main1 extends Frame implements ActionListener {
```

```
    CardLayout card;
```

```
    Panel mainPanel;
```

```
    Main1() {
```

```
        setTitle("Java AWT Programming Lab - All Questions");
```

```
        setSize(700, 600);
```

```
        setLayout(new BorderLayout());
```

```
        // Main menu
```

```
        Panel menu = new Panel(new GridLayout(4, 2, 10, 10));
```

```
        String[] buttons = {
```

```
            "Q1 Applet Life Cycle",
```

```
            "Q2 Graphics",
```

```
            "Q3 Calculator Parameters",
```

```
            "Q4 Student Profile",
```

```
            "Q5 Grid Calculator",
```

```
            "Q6 GridBag Registration",
```

```
            "Q7 BorderLayout Report",
```

```
            "Q8 CardLayout"
```

```
        };
```

```
        for (String s : buttons) {
```

```
            Button b = new Button(s);
```

```
            b.addActionListener(this);
```

```
            menu.add(b);
```

```
        }
```

```
        add(menu, BorderLayout.NORTH);
```

```
card = new CardLayout();
mainPanel = new Panel(card);
```

```
mainPanel.add(q1(), "Q1 Applet Life Cycle");
mainPanel.add(q2(), "Q2 Graphics");
mainPanel.add(q3(), "Q3 Calculator Parameters");
mainPanel.add(q4(), "Q4 Student Profile");
mainPanel.add(q5(), "Q5 Grid Calculator");
mainPanel.add(q6(), "Q6 GridBag Registration");
mainPanel.add(q7(), "Q7 BorderLayout Report");
mainPanel.add(q8(), "Q8 CardLayout");
```

```
add(mainPanel, BorderLayout.CENTER);
```

```
addWindowListener(new WindowAdapter() {
    public void windowClosing(WindowEvent e) {
        System.exit(0);
    }
});
```

```
setVisible(true);
}
```

```
// =====
// Q1 - APPLETT LIFE CYCLE
// =====
```

```
Panel q1() {
```

```
    Panel p = new Panel(new BorderLayout());
```

```
    TextArea t = new TextArea();
```

```
    t.setText(
        "Q1 - APPLETT LIFE CYCLE\n\n" +
        "init()  -> Initializes the applet\n" +
        "start()  -> Starts the applet\n" +
        "paint()  -> Displays output\n" +
        "stop()   -> Stops the applet\n" +
        "destroy() -> Destroys the applet\n\n" +
```

```

        "Execution Sequence:\n" +
        "init() -> start() -> paint() -> stop() -> destroy()"
    );

    p.add(t, BorderLayout.CENTER);

    return p;
}

// =====
// Q2 - GRAPHICS
// =====

Panel q2() {

    return new GraphicsPanel();
}

class GraphicsPanel extends Panel {

    public void paint(Graphics g) {

        // Sky
        g.setColor(Color.cyan);
        g.fillRect(0, 0, 700, 400);

        // Sun
        g.setColor(Color.yellow);
        g.fillOval(50, 50, 80, 80);

        // Clouds
        g.setColor(Color.white);
        g.fillOval(180, 60, 80, 40);
        g.fillOval(220, 40, 90, 60);
        g.fillOval(270, 60, 80, 40);

        // Ground
        g.setColor(Color.green);
        g.fillRect(0, 320, 700, 150);
    }
}

```

```
// Road
g.setColor(Color.gray);
g.fillRect(280, 320, 140, 150);

// Road lines
g.setColor(Color.white);
g.drawLine(350, 330, 350, 350);
g.drawLine(350, 365, 350, 385);
g.drawLine(350, 400, 350, 420);
g.drawLine(350, 435, 350, 455);

// House
g.setColor(Color.orange);
g.fillRect(100, 190, 180, 130);

// Roof
g.setColor(Color.red);

int x[] = {80, 190, 300};
int y[] = {190, 100, 190};

g.fillPolygon(x, y, 3);

// Door
g.setColor(Color.darkGray);
g.fillRect(170, 250, 45, 70);

// Windows
g.setColor(Color.white);
g.fillRect(120, 220, 35, 35);
g.fillRect(225, 220, 35, 35);

// Tree 1
g.setColor(Color.green);
g.fillOval(450, 190, 80, 100);

g.setColor(Color.darkGray);
g.fillRect(480, 270, 20, 70);

// Tree 2
```

```

        g.setColor(Color.green);
        g.fillOval(560, 210, 80, 100);

        g.setColor(Color.darkGray);
        g.fillRect(590, 290, 20, 60);
    }
}

// =====
// Q3 - CALCULATOR USING PARAMETERS
// =====

Panel q3() {

    Panel p = new Panel(new GridLayout(5, 2, 10, 10));

    TextField n1 = new TextField();
    TextField n2 = new TextField();
    TextField op = new TextField();
    TextField result = new TextField();

    Button calculate = new Button("Calculate");

    p.add(new Label("Number 1:"));
    p.add(n1);

    p.add(new Label("Number 2:"));
    p.add(n2);

    p.add(new Label("Operator (+,-,*,/):"));
    p.add(op);

    p.add(calculate);

    p.add(new Label("Result:"));
    p.add(result);

    calculate.addActionListener(new ActionListener() {

        public void actionPerformed(ActionEvent e) {

```

```

try {

    double a = Double.parseDouble(n1.getText());
    double b = Double.parseDouble(n2.getText());
    String operator = op.getText();

    double r = 0;

    if (operator.equals("+"))
        r = a + b;
    else if (operator.equals("-"))
        r = a - b;
    else if (operator.equals("*"))
        r = a * b;
    else if (operator.equals("/"))
        r = a / b;
    else {
        result.setText("Invalid Operator");
        return;
    }

    result.setText(String.valueOf(r));

} catch (Exception ex) {
    result.setText("Invalid Input");
}
});

return p;
}

// =====
// Q4 - STUDENT PROFILE
// =====

Panel q4() {

    Panel p = new Panel(new FlowLayout());

```

```
TextField name = new TextField(20);
```

```
TextField reg = new TextField(15);
```

```
CheckboxGroup genderGroup = new CheckboxGroup();
```

```
Checkbox male =  
    new Checkbox("Male", genderGroup, false);
```

```
Checkbox female =  
    new Checkbox("Female", genderGroup, false);
```

```
Choice department = new Choice();  
department.add("CSE");  
department.add("ECE");  
department.add("EEE");  
department.add("IT");  
department.add("MECH");
```

```
Choice year = new Choice();  
year.add("1");  
year.add("2");  
year.add("3");  
year.add("4");
```

```
List skills = new List(4, true);  
skills.add("Java");  
skills.add("Python");  
skills.add("C");  
skills.add("C++");  
skills.add("HTML");
```

```
TextArea address = new TextArea(5, 30);
```

```
Button submit = new Button("Submit");
```

```
p.add(new Label("Name:"));  
p.add(name);
```

```
p.add(new Label("Register Number:"));
```

```

p.add(reg);

p.add(new Label("Gender:"));
p.add(male);
p.add(female);

p.add(new Label("Department:"));
p.add(department);

p.add(new Label("Year:"));
p.add(year);

p.add(new Label("Skills:"));
p.add(skills);

p.add(new Label("Address:"));
p.add(address);

p.add(submit);

submit.addActionListener(new ActionListener() {

    public void actionPerformed(ActionEvent e) {

        String g = "";

        if (genderGroup.getSelectedCheckbox() != null)
            g = genderGroup.getSelectedCheckbox().getLabel();

        System.out.println("----- STUDENT PROFILE -----");
        System.out.println("Name: " + name.getText());
        System.out.println("Register Number: " + reg.getText());
        System.out.println("Gender: " + g);
        System.out.println(
            "Department: " +
            department.getSelectedItem()
        );
        System.out.println(
            "Year: " +
            year.getSelectedItem()

```



```

        );
        System.out.println(
            "Address: " +
            address.getText()
        );
    }
});

return p;
}

// =====
// Q5 - GRIDLAYOUT CALCULATOR
// =====

Panel q5() {

    Panel p = new Panel(new BorderLayout());

    TextField display = new TextField();
    display.setEditable(false);

    p.add(display, BorderLayout.NORTH);

    Panel buttons = new Panel();
    buttons.setLayout(new GridLayout(4, 4, 5, 5));

    String[] b = {
        "7", "8", "9", "+",
        "4", "5", "6", "-",
        "1", "2", "3", "*",
        "0", "=", "C", "/"
    };

    final double[] num1 = new double[1];
    final double[] num2 = new double[1];
    final char[] operator = new char[1];

    for (String s : b) {

```

```

Button button = new Button(s);
buttons.add(button);

button.addActionListener(new ActionListener() {

    public void actionPerformed(ActionEvent e) {

        String value = e.getActionCommand();

        if (value.equals("C")) {

            display.setText("");

        } else if (
            value.equals("+") ||
            value.equals("-") ||
            value.equals("*") ||
            value.equals("/")
        ) {

            num1[0] =
                Double.parseDouble(display.getText());

            operator[0] = value.charAt(0);

            display.setText("");

        } else if (value.equals("=")) {

            num2[0] =
                Double.parseDouble(display.getText());

            double result = 0;

            switch (operator[0]) {

                case '+':
                    result = num1[0] + num2[0];
                    break;

```

```

        case '-':
            result = num1[0] - num2[0];
            break;

        case '*':
            result = num1[0] * num2[0];
            break;

        case '/':
            result = num1[0] / num2[0];
            break;
    }

    display.setText(
        String.valueOf(result)
    );

} else {

    display.setText(
        display.getText() + value
    );
}

});
}

p.add(buttons, BorderLayout.CENTER);

return p;
}

// =====
// Q6 - GRIDBAGLAYOUT REGISTRATION
// =====

Panel q6() {

    Panel p = new Panel(new GridBagLayout());

```

```
GridBagConstraints c =
    new GridBagConstraints();

c.insets = new Insets(5, 5, 5, 5);
c.fill = GridBagConstraints.HORIZONTAL;

TextField name = new TextField();
TextField reg = new TextField();
TextField email = new TextField();

Choice dept = new Choice();
dept.add("CSE");
dept.add("ECE");
dept.add("EEE");
dept.add("IT");

Choice year = new Choice();
year.add("1");
year.add("2");
year.add("3");
year.add("4");

CheckboxGroup gender =
    new CheckboxGroup();

Checkbox male =
    new Checkbox("Male", gender, false);

Checkbox female =
    new Checkbox("Female", gender, false);

Button submit = new Button("Submit");
Button reset = new Button("Reset");

addGB(p, new Label("Name:"), 0, 0, c);
addGB(p, name, 1, 0, c);

addGB(p, new Label("Register Number:"), 0, 1, c);
addGB(p, reg, 1, 1, c);
```

```
addGB(p, new Label("Department:"), 0, 2, c);
addGB(p, dept, 1, 2, c);
```

```
addGB(p, new Label("Email:"), 0, 3, c);
addGB(p, email, 1, 3, c);
```

```
addGB(p, new Label("Gender:"), 0, 4, c);
```

```
Panel gp = new Panel();
gp.add(male);
gp.add(female);
```

```
addGB(p, gp, 1, 4, c);
```

```
addGB(p, new Label("Year:"), 0, 5, c);
addGB(p, year, 1, 5, c);
```

```
addGB(p, submit, 0, 6, c);
addGB(p, reset, 1, 6, c);
```

```
submit.addActionListener(new ActionListener() {

    public void actionPerformed(ActionEvent e) {

        String g = "";

        if (gender.getSelectedCheckbox() != null)
            g = gender.getSelectedCheckbox().getLabel();

        System.out.println(
            "Name: " + name.getText()
        );

        System.out.println(
            "Register Number: " + reg.getText()
        );

        System.out.println(
            "Department: " +
            dept.getSelectedItem()
        );
    }
});
```

```

    );

    System.out.println(
        "Email: " + email.getText()
    );

    System.out.println("Gender: " + g);

    System.out.println(
        "Year: " + year.getSelectedItem()
    );
}
});

reset.addActionListener(new ActionListener() {

    public void actionPerformed(ActionEvent e) {

        name.setText("");
        reg.setText("");
        email.setText("");
    }
});

return p;
}

void addGB(
    Panel p,
    Component comp,
    int x,
    int y,
    GridBagConstraints c
) {

    c.gridx = x;
    c.gridy = y;

    p.add(comp, c);
}

```

```
// =====  
// Q7 - BORDERLAYOUT REPORT  
// =====
```

```
Panel q7() {
```

```
    Panel main = new Panel(new BorderLayout(10, 10));
```

```
    Label header =  
        new Label("STUDENT REPORT", Label.CENTER);
```

```
    main.add(header, BorderLayout.NORTH);
```

```
    Panel center =  
        new Panel(new GridLayout(6, 2, 5, 5));
```

```
    TextField name = new TextField();  
    TextField reg = new TextField();
```

```
    Choice gender = new Choice();  
    gender.add("Male");  
    gender.add("Female");
```

```
    Choice dept = new Choice();  
    dept.add("CSE");  
    dept.add("ECE");  
    dept.add("EEE");  
    dept.add("IT");
```

```
    Choice year = new Choice();  
    year.add("1");  
    year.add("2");  
    year.add("3");  
    year.add("4");
```

```
    TextField skills = new TextField();
```

```
    center.add(new Label("Name:"));  
    center.add(name);
```

```
center.add(new Label("Register Number:"));
center.add(reg);

center.add(new Label("Gender:"));
center.add(gender);

center.add(new Label("Department:"));
center.add(dept);

center.add(new Label("Year:"));
center.add(year);

center.add(new Label("Skills:"));
center.add(skills);

main.add(center, BorderLayout.CENTER);

TextArea report = new TextArea();

main.add(report, BorderLayout.EAST);

Panel bottom = new Panel();

Button generate =
    new Button("Generate Report");

Button clear =
    new Button("Clear");

bottom.add(generate);
bottom.add(clear);

main.add(bottom, BorderLayout.SOUTH);

generate.addActionListener(new ActionListener() {

    public void actionPerformed(ActionEvent e) {

        report.setText(
```



```

        "STUDENT REPORT\n\n" +
        "Name: " + name.getText() + "\n" +
        "Register Number: " + reg.getText() + "\n" +
        "Gender: " + gender.getSelectedItem() + "\n" +
        "Department: " + dept.getSelectedItem() + "\n" +
        "Year: " + year.getSelectedItem() + "\n" +
        "Skills: " + skills.getText()
    );
}
});

clear.addActionListener(new ActionListener() {

    public void actionPerformed(ActionEvent e) {

        name.setText("");
        reg.setText("");
        skills.setText("");
        report.setText("");
    }
});

return main;
}

// =====
// Q8 - CARDLAYOUT
// =====

Panel q8() {

    Panel main = new Panel(new BorderLayout());

    CardLayout cl = new CardLayout();
    Panel cards = new Panel(cl);

    // CARD 1
    Panel student = new Panel(
        new GridLayout(3, 2, 10, 10)
    );

```

```
TextField name = new TextField();
TextField reg = new TextField();
```

```
Choice dept = new Choice();
```

```
dept.add("CSE");
dept.add("ECE");
dept.add("EEE");
dept.add("IT");
```

```
student.add(new Label("Student Name:"));
student.add(name);
```

```
student.add(new Label("Register Number:"));
student.add(reg);
```

```
student.add(new Label("Department:"));
student.add(dept);
```

```
// CARD 2
```

```
Panel academic = new Panel(
    new GridLayout(2, 2, 10, 10)
);
```

```
Choice year = new Choice();
```

```
year.add("1");
year.add("2");
year.add("3");
year.add("4");
```

```
TextField marks = new TextField();
```

```
academic.add(new Label("Year:"));
academic.add(year);
```

```
academic.add(new Label("Marks:"));
academic.add(marks);
```

```
// CARD 3
TextArea confirmation =
    new TextArea();

cards.add(student, "student");
cards.add(academic, "academic");
cards.add(confirmation, "confirmation");

main.add(cards, BorderLayout.CENTER);

Panel buttons = new Panel();

Button previous =
    new Button("Previous");

Button next =
    new Button("Next");

Button submit =
    new Button("Submit");

buttons.add(previous);
buttons.add(next);
buttons.add(submit);

main.add(buttons, BorderLayout.SOUTH);

next.addActionListener(new ActionListener() {

    public void actionPerformed(ActionEvent e) {

        cl.next(cards);
    }
});

previous.addActionListener(new ActionListener() {

    public void actionPerformed(ActionEvent e) {

        cl.previous(cards);
    }
});
```

```

    }
});

submit.addActionListener(new ActionListener() {

    public void actionPerformed(ActionEvent e) {

        confirmation.setText(
            "CONFIRMATION / REPORT\n\n" +
            "Name: " + name.getText() + "\n" +
            "Register Number: " +
            reg.getText() + "\n" +
            "Department: " +
            dept.getSelectedItem() + "\n" +
            "Year: " +
            year.getSelectedItem() + "\n" +
            "Marks: " + marks.getText()
        );

        cl.show(cards, "confirmation");
    }
});

return main;
}

// =====
// MAIN EVENT HANDLING
// =====

public void actionPerformed(ActionEvent e) {

    String command = e.getActionCommand();

    card.show(mainPanel, command);
}

// =====
// MAIN METHOD
// =====

```

```
public static void main(String[] args) {  
    new Main1();  
}
```

Q1 Applet Life Cycle	Q2 Graphics
Q3 Calculator Parameters	Q4 Student Profile
Q5 Grid Calculator	Q6 GridBag Registration
Q7 BorderLayout Report	Q8 CardLayout
Q1 - APPLET LIFE CYCLE init() -> Initializes the applet start() -> Starts the applet paint() -> Displays output stop() -> Stops the applet destroy() -> Destroys the applet Execution Sequence: init() -> start() -> paint() -> stop() -> destroy()	



Q3 Calculator Parameters	Q4 Student Profile
Q5 Grid Calculator	Q6 GridBag Registration
Q7 BorderLayout Report	Q8 CardLayout
Number 1:	
Number 2:	
Operator (+,-,*,/):	
Calculate	Result:

Q1 Applet Life Cycle		Q2 Graphics	
Q3 Calculator Parameters		Q4 Student Profile	
Q5 Grid Calculator		Q6 GridBag Registration	
Q7 BorderLayout Report		Q8 CardLayout	
7	8	9	+
4	5	6	-
1	2	3	*
0	=	C	/

Q1 Applet Life Cycle	Q2 Graphics
Q3 Calculator Parameters	Q4 Student Profile
Q5 Grid Calculator	Q6 GridBag Registration
Q7 BorderLayout Report	Q8 CardLayout

Name:

Register Number:

Department: 

Email:

Gender: ☐ Male ☐ Female

Year: 

STUDENT REPORT

Name:

Register Number:

Gender:

Department:

Year:

Skills:

Generate Report

Clear

Student Name:

Register Number:

Department:

Previous

Next

Submit

Q1 Applet Life Cycle	Q2 Graphics
Q3 Calculator Parameters	Q4 Student Profile
Q5 Grid Calculator	Q6 GridBag Registration
Q7 BorderLayout Report	Q8 CardLayout

Name: Register Number: Gender: ☐ Male ☐ Female

Department:

Year:

Skills:

Address: